Lesson notes · for you

Switches that decide

IoT · Level 2 · Lesson 2

Materials
Paper and a pen; No kit? The logic lab on a phone or computer; Optional: two AA batteries, an LED with a 100 Ω resistor, kitchen foil and two paper clips

Learning goals

- Explain AND, OR and NOT with switches and lights.
- Fill in a truth table for a simple rule.
- Design the logic for a real device, like a fan or an alarm.

Activity: Truth table detective

1. Look around your home, your school or the bus: a fridge light, a torch, a phone screen, a kettle, a bus bell.
2. Write when each one acts, as an ordinary sentence: “The fridge light is on when the door is open.”
3. Name the inputs, and choose the gate: all needed (AND), any one enough (OR), or the opposite (NOT).
4. List every row, with 0 and 1 for each input (4 rows for two inputs), and write the output for each.
5. Read each row aloud. If the device would do something silly, change the gate and check the table again.

No kit needed. If you build with batteries: batteries only, press the paper clips onto foil strips to make switches, and never join the battery's + and - with foil alone.

Discussion

- Which device's rule was hardest to write, and why?
- Where would a mistake in a rule waste money or be unsafe?
- Why is it worth writing the truth table before you build?

Check yourself

- Write the truth table for A AND NOT B.

Try it this week
Truth table detective: Find three devices at home or at school that decide something: a fridge light, a torch, a phone screen, a kettle, a bus bell. For each, write its rule with AND, OR or NOT, and make its truth table with every row. Then look for a row where the rule would do something silly, and fix it. If you have two AA batteries, an LED with a 100 Ω resistor, kitchen foil and two paper clips, build AND (clips in series) and OR (clips in parallel), and test every row.

Worksheet

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Name _____ Class _____ Date _____

1 **A pump fills the school tank from a well. It must run when the tank is low, but never when the well is dry, or it burns out. Which rule is right?**

◯ Tank low OR well dry ◯ Tank low AND NOT well dry ◯ Tank low AND well dry

2 **On a bus, any passenger's button rings the driver's bell. So the buttons must be wired as AND, in series.**

◯ True ◯ False

3 **Complete the truth-table rows.**

For AND, 1 and 0 give _____. For OR, 1 and 1 give _____.

Word bank: 11 · 0 · 2 · 1

4 **Match each device's rule to its gate.**

The bus bell rings if any passenger presses a button	• • • •	AND
A blender runs only if the lid is on and you press Start	• • • •	XOR
A night light is on when it is not bright	• • • •	OR
A stair light changes when either switch is flipped	• • • •	NOT

5 **Kofi programs his classroom fan as: hot OR someone in the room. What goes wrong?**

◯ It runs only when it's hot and someone is in ◯ It never runs when the room is full

◯ It runs in an empty room whenever it's hot

✎ **Think of a device you use every day. Write its rule with AND, OR or NOT, and say what would go wrong with the wrong gate.**

Check your answers

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1 **Tank low AND NOT well dry**

The pump should run only when the tank needs water AND the well is NOT dry.

2 **False**

One button is enough, so it's OR: the buttons are in parallel, each giving the current its own path.

3 **For AND, 1 and 0 give 0. For OR, 1 and 1 give 1.**

AND needs every input to be 1, so 1 and 0 give 0.

4 **The bus bell rings if any passenger presses a button — OR**
A blender runs only if the lid is on and you press Start — AND
A night light is on when it is not bright — NOT
A stair light changes when either switch is flipped — XOR

Listen for the meaning: any one is enough (OR); all are needed (AND); the opposite (NOT); exactly one different (XOR).

5 **It runs in an empty room whenever it's hot**

OR needs just one input, so the row "hot, nobody in" gives 1 and the fan spins in an empty room, wasting power.